



Growth and metabolic performance of black soldier fly larvae grown on low and high-quality substrates

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ABSTRACT

We have measured growth and respiration in black soldier fly (BSF) larvae fed with mixtures of a low-quality substrate (degassed sludge, DS), and a high-quality substrate (chicken feed, CF) in order to elucidate how substrate quality affect larval metabolism and feed conversion into new biomass. The BSF larvae grew faster and became larger the higher the content of CF was in the substrate. Growth followed a sigmoidal curve from where the specific growth rate was determined and compared to measured specific respiration rates, in order to estimate costs of growth, maintenance metabolism, rates of feed assimilation, and net growth efficiency. Specific feed assimilation rates were similar on all substrate mixtures. Maximal specific growth rates were also affected only little unless the larvae were grown in pure DS. In contrast, the cost of growth and the maintenance metabolism were larger the higher the proportion of DS was. High specific growth rates were, in addition, sustained for shorter periods of the time the more DS was included in the substrate mixtures. In effect, higher proportions of the assimilated feed were spent on respiratory purposes instead of being converted into larval biomass and the net growth efficiency decreased the more DS was included in the substrate mixtures. We conclude that substrate quality may affect the conversion of feed into new biomass via alterations of the metabolic performance of BSF larvae and thereby the overall performance of BSF larval cultures.

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1. Introduction

The black soldier fly (BSF), *Hermetia illucens* receives substantial attention for use in waste management, and for feed and food production (Gold et al., 2018; Bosch et al., 2019). BSF larvae have been grown on a variety of substrates, ranging from organic wastes and different types of sludge over various agricultural by-products to commercial animal feed blends (Diener et al., 2009; Oonincx et al., 2015; Rehman et al., 2017; Abduh et al., 2018; Liu et al., 2018; Meneguz et al., 2018; Ganda et al., 2019; Lalander et al., 2019; Palma et al., 2019; Gold et al., 2020). The efficiency by which BSF larvae utilise different substrates for growth is highly variable (Bosch et al., 2019) and may be related to the quality of the substrate. Earlier results have shown that degassed (digested) sludge from anaerobic digesters is an example of a low-quality substrate for BSF larvae (Lalander et al., 2019). Although rich in proteins and other constituents of nutritional value, growth of BSF larvae on degassed sludge is very slow, the larvae remain small and do not complete their development into prepupae, and feed conversion into new biomass is low (Lalander et al., 2019). Commercial

chicken feed blends are, on the other hand, excellent substrates for BSF larvae, supporting rapid growth and larval development (Diener et al., 2009; Liu et al., 2017a; Lalander et al., 2019). Some variations in the performance of BSF larvae across substrates have been explained by the ratio of proteins to carbohydrates (Barragán-Fonseca et al., 2019; Gold et al., 2020), quality of the macromolecular components (Barragán-Fonseca et al., 2018), moisture content (Cheng et al., 2017), and indigestible components (Gold et al., 2020).

Feed conversion efficiency in BSF larvae is most often evaluated over the entire larval growth phase. BSF larvae have been able to convert 2–16% of different substrates into new biomass (Diener et al., 2009; Rehman et al., 2017; Abduh et al., 2018; Ganda et al., 2019; Lalander et al., 2019), 14–24% of ingested feed into new biomass (Oonincx et al., 2015), or 4–38% of digested feed into new biomass (Diener et al., 2009; Abduh et al., 2018; Liu et al., 2018; Meneguz et al., 2018; Palma et al., 2019; Gold et al., 2020) when measurements are based on dry weights (DW). Substrates for BSF larvae are moist and contain active microorganisms (Palma et al., 2019). At certain conditions, microbial substrate consumption may even exceed that of the larvae (Bekker et al., submitted for publication), and therefore feed conversion efficiency does not solely reflect the actual performance of the

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BSF larvae. More insight into the metabolic performance of BSF larvae is provided by the net growth efficiency, *NGE* (or net energy of growth), as this describes the ratio between biomass gain and feed assimilation (see e.g. Chaplin and Chaplin, 1981). Hence, *NGE* is an efficiency measure that takes only the feed components actually assimilated by the BSF larvae into consideration and defines the upper limit of feed conversion efficiency.

Information on larval *NGE* across substrates of low and high quality will be highly useful for optimization of BSF substrate compositions and production processes and enable different waste- and by-products to be utilised most efficiently as feed for BSF larvae. The aim of this study is to quantify and compare the metabolic performance of BSF larvae growing on mixtures (0–100%) of a low-quality substrate (degassed sludge) and a high-quality substrate (commercial chicken feed) in terms of *NGE*, combined with estimates of costs of growth and maintenance, as well as feed assimilation, respiration, and growth. The purpose is to elucidate how and to which extent the metabolic performance of BSF larvae may be affected by the quality and the nature of the growth substrate, and how this translates into variability of *NGE*.

2. Methods

2.1. Cultivation of parental black soldier flies and nursery

Domesticated black soldier flies (BSF), *Hermetia illucens* were supplied by the Danish company, Entomass ApS, which is a commercial producer of BSF larvae. A breeding culture of BSF was established in the laboratory in a temperature-controlled chamber at 28 °C, a relative humidity of 67%, and a 12:12 h light:dark period. Larvae were reared on broiler chicken feed (CF) containing 70% water in a cylindrical bucket (diameter 30 cm, height 35 cm, initial depth of feed 10 cm). Additional CF was added regularly. From the 3rd generation of adult flies reared in the laboratory, 0.7 g eggs were harvested from an egg trap made of cardboard placed above a mixture of decaying banana and apple. The eggs were transferred to a nursery made from a cylindrical bucket (10 cm in diameter, 13 cm in height, initial depth of CF, 5 cm). After 10 days of incubation, the neonate larvae had reached a size where they could be counted and handled without damage.

2.2. Substrates

Substrates for the BSF larvae were prepared from mixtures (0–100%) of degassed sludge (DS) collected at a local biogas facility (Nature Energy, Vaarst, Denmark) producing biogas from cattle and pig manure supplemented with industrial organic waste, and a commercial chicken feed (CF) blend (Paco Start 19, DLG, Denmark). Before use, DS was sieved through a 1 mm sieve to remove larger particles and subsequently centrifuged at 1000 g for 5 min. after which the supernatant was discharged in order to increase the dry matter content. Water was added until saturation after the substrates had been mixed. Lipid (measured by NMR), carbohydrate (measured as monosaccharides on an HPLC after enzymatic digestion), protein (measured by the Dumas method) and ash contents (measured gravimetrically after combustion in an ashing furnace) in DS and CF were quantified by an accredited laboratory for nutritional and chemical analyses of foods (ALS, Humlebæk, Denmark).

2.3. Cultivation of BSF larvae

BSF larvae were grown in cylindrical plastic buckets (10 cm in diameter, 13 cm in height) containing 50 g DW of water-saturated substrate mixtures. Larval growth on each substrate

mixture was examined in 4 replicate cultures. Fifty neonate larvae, 10 days of age, were distributed into each of the buckets on Day 0 of the cultivation trials. The buckets were incubated under the same conditions as the breeding culture. Each culture was supplemented with another 50 g (DW) of the same water-saturated substrate mixture as originally used on Day 6 (post inoculation) except for the cultures grown on pure sludge, which were supplemented with only 16.7 g (DW) of water-saturated DS. Prepupae were sampled from a petri dish placed on top of the substrate on a daily basis and stored frozen for further analysis of DW and organic content.

2.4. CO₂ production rate

Respiration rates, quantified as the rate of CO₂ production per individual, Q_{CO_2} were measured on a daily basis on 6 randomly selected larvae from each bucket. Measurements were carried out as quickly and gently as possible. Immediately after larvae had been collected, they were transferred to a 50 mL plastic centrifugation tube placed in an aluminium heating block (Hettich Benelux model MHR-23) at 28 °C. A pre-calibrated PS-3208 wireless CO₂ Sensor (Pasco, Roseville, California) was inserted into the top of the tube and the CO₂ concentration in the tube was recorded at a frequency of 1 Hz. The linear increase in CO₂ concentration recorded for a period of 3 min, starting 2 min after the tube had been closed, was used as measure of Q_{CO_2} .

2.5. Wet weight, dry weight and organic fraction

After CO₂ production rates had been quantified, the wet weight of the 6 randomly selected larvae (including gut content) was measured on an electronic scale before they were returned alive to their original bucket.

Dry weight (DW) was measured at 3-day intervals on 5 BSF larvae randomly collected from cultures grown under comparable conditions and in parallel to the cultures used for growth experiments, in the same substrate mixtures and under the same conditions. The larvae were killed by freezing and dried for 24 h at 105 °C. It was assumed that the dry matter content of larvae in the growth experiments resembled these measurements.

Wet weight and dry weight were measured also on the prepupae collected on top of all cultures. The dried prepupae were then transferred to a furnace at 550 °C for 5 h and the weight of the remaining ash was measured. The mass difference between dry weight and ash was used to evaluate the organic fraction of the prepupae.

2.6. Size, growth rate and specific growth rate

In order to describe the increase in the size of the larvae and their specific growth rate during the growth phase, larval dry weight, X was described by a logistic function

$$X = \frac{X_{max}}{1 + \left(\frac{X_{max}-X_0}{X_0}\right)e^{-\mu_{max}t}} \quad (1)$$

where X_{max} is the maximal dry weight of larvae obtained during the growth phase, and μ_{max} is their maximal specific growth rate. Eq. (1) was fitted to the measured dry weights of BSF larvae by finding the combination of X_{max} and μ_{max} , which resulted in the smallest average mean error between measured and modelled dry weights in the period from Day 1 and until half of the larvae developed into prepupae. The growth rate, r_X was then determined from the first-order derivative of Eq. (1)

$$r_X = \frac{dX}{dt} = \mu_{max}X \left(1 - \frac{X}{X_{max}}\right) \quad (2)$$

Similar approaches have been used to estimate growth rates e.g. in chickens (Duan-yai et al., 1999). The specific growth rate was finally estimated as

$$\mu = \frac{r_X}{X} = \mu_{\max} \left(1 - \frac{X}{X_{\max}} \right) \quad (3)$$

until the BSF larvae developed into prepupae.

2.7. Specific rate of assimilation, cost of growth and maintenance coefficient

If growth is balanced, the assimilated feed will either be incorporated into the biomass or metabolised in order to generate energy, and the specific assimilation rate was estimated as the sum of the specific growth rate, μ and the specific respiration rate, q in terms of carbon equivalents

$$a = \mu + q \quad (4)$$

Respiration supplies energy for growth and for maintenance purposes and was therefore split into a growth-dependent and a growth-independent term (see e.g. Pirt, 1982)

$$q = Y\mu + m \quad (5)$$

where Y represents the cost of growth and m is the specific rate of respiration needed to support the maintenance requirements. Y will be constant if growth is balanced. If m is independent of the size and age of the larvae, which may be the case in small, immature animals (Riisgård, 1998), q and μ will describe a linear relationship during the growth phase. Y and m were estimated from the slope and the intercept with the y-axis, respectively of plots of q_{CO_2} vs. μ . The maximal q_{CO_2} , $q_{CO_2, \max}$ was estimated by substituting μ by the maximal specific growth rate, $\mu_{\max}$.

2.8. Net growth efficiency

The net growth efficiency, NGE can be expressed as the ratio between specific rates of growth, μ and feed assimilation, a

$$NGE = \frac{\mu}{a} \quad (6)$$

When the larvae reach the prepupae, state they stop feeding (Gligorescu et al., 2019) and cover their energetic needs by recycling their own biomass. The mass of the prepupae decreases (Liu et al., 2017a; Barragán-Fonseca et al., 2018) and growth will no longer be balanced. Until larvae become prepupae, NGE was evaluated by Eq. (7), derived by combining Eqs. (4) and (6)

$$NGE = \frac{\mu}{\mu + q} \quad (7)$$

If Eq. (5) is inserted into Eq. (7),

$$NGE = \frac{\mu}{\mu + Y\mu + m} = \frac{1}{1 + Y + \frac{m}{\mu}} \quad (8)$$

it is seen that NGE is a function of the cost of growth, Y and the ratio between specific rates of maintenance costs, m and growth, μ . The maximal NGE , $NGE_{\max}$ was estimated by substituting μ by the maximal specific growth rate, $\mu_{\max}$. The average NGE from inoculation until the larvae formed prepupae, NGE_{avg} was evaluated as the ratio between the weight gained by the BSF larvae and the integrated amount of feed assimilated during the growth phase.

2.9. Conversion factors

In order to compare measurements of growth and CO_2 production, all measurements were expressed as specific rates based on the mass of carbon in the larvae and in CO_2 . The specific CO_2

production rate, q_{CO_2} , was calculated by multiplying the larval CO_2 production rate, Q_{CO_2} by the mass-fraction of carbon in CO_2 and dividing by the mass of carbon in one larva. We assumed that carbon made up 49% of the organic fraction of the larvae, which is the carbon content in biomass of average elementary composition (Roels, 1980). The organic fraction of the larvae was estimated from measurements of the dry weight of the larvae and measurements of the organic fraction of prepupae reared on each substrate mixture.

In order to express Y , m and NGE in terms of energy, we considered that the combustion heat of biomass of average elementary composition is $560 \text{ kJ C-mol}^{-1}$, corresponding to 22.7 kJ g^{-1} (Roels, 1980). It was assumed, that also the feed components actually assimilated by the BSF larvae had an average elementary composition similar to the BSF larvae.

2.10. Statistical analysis

Results from 4 replicate cultures on all substrate mixtures were compared in terms of a null hypothesis of equal larval performance on all substrate mixtures. Larval performances were considered significantly different when the null hypothesis was rejected by ANOVA analysis and t-tests at 5% probability level.

3. Results

3.1. Growth and development

Chicken feed (CF) had the highest content of carbohydrates while degassed sludge (DS) had highest contents of proteins and inorganic components. Both substrates had low lipid contents (Table 1). The BSF larvae grew on both substrates and on mixtures of the two, though no larvae reached the prepupal stage within 30 days when grown solely on DS. The higher the content of CF in the substrate mixtures, the larger became the proportions of individuals that developed into prepupae (Fig. 1, left panels). With 25% CF in the substrate mixtures, 32% of the larvae developed into prepupae within 30 days, increasing to 80–89% on the substrate mixtures with highest CF contents (Table 1). DS also prolonged the larval stage of the BSF larvae. The first prepupae from larvae grown on 100% CF appeared on Day 13 (post inoculation) and half of the prepupae had formed on Day 17, 27 days of age. Larvae grown on 75% DS began to develop into prepupae on Day 17 and half of the prepupae had formed on Day 23, 33 days of age. Inclusion of 75% DS in the substrate mixtures thus prolonged larval development by 6 days. The dry weight content of the larvae increased during the growth phase (Supplementary information, Fig. S1) and the average dry weight content of the larvae and the organic fraction of the dry weight were highest in substrate mixtures rich in CF (Supplementary information, Fig. S2).

The maximal size of larvae, $X_{\max}$ and the maximal specific growth rate, $\mu_{\max}$, estimated from Eq. (1) was only 6.9 mg and 0.22 day^{-1} , respectively, for larvae grown on DS (Table 1). Larvae grew significantly larger on substrates containing CF and the higher the CF content, the larger became the larvae ($p < 0.05$). $X_{\max}$ was almost 3 times larger for larvae grown on pure CF (94 mg DW) compared to larvae grown on 75% DS (38 mg DW). In contrast, $\mu_{\max}$ were similar (0.54 – 0.60 day^{-1}) for larvae grown on 0–75% DS (for larvae grown on 25–75% sludge, differences in $\mu_{\max}$ were not significant, $p < 0.05$). The specific growth rate remained close to its maximal value for about 7 days (14 days for larvae grown in DS), in which period growth was exponential (Supplementary information, Fig. S3). The growth rate of the larvae, r_X (Eq. (2)) peaked on Day 8 on all substrate mixtures, except for 100% DS in which r_X did not peak until Day 11 (Fig. 1, left panels). The period from

Table 1

Nutritional composition of substrate mixtures made from degassed sludge (DS) and chicken feed (CF) and performance of BSF larvae. Fraction of larvae developing into prepupae within 30 days, f_p . Time until half the prepupae had developed, t_p . Maximal size of larvae, X_{max} and maximum specific growth rate, μ_{max} (Eq. (1)). Cost of growth, Y , maintenance coefficient, m , and coefficient of determination, r^2 (Eq. (5)). Maximal specific CO₂ production rate, $q_{CO_2,max}$ (Eq. (5)). Maximal specific assimilation rate, a_{max} (Eq. (4)). Maximal and average net growth efficiency, NGE_{max} (Eq. (8)) and NGE_{avg} , respectively.

Substrate mixture		100% DS 0% CF	75% DS 25% CF	50% DS 50% CF	25% DS 75% CF	0% DS 100% CF
Parameter/variable	Unit	Substrate composition				
Degassed sludge	% DW	100	75	50	25	0
Chicken feed	% DW	0	25	50	75	100
Lipid	% DW	2.7	3.0	3.2	3.5	3.7
Carbohydrate	% DW	40.0	47.3	54.7	62.0	69.3
Protein	% DW	30.6	28.2	25.8	23.3	20.9
Ash	% DW	26.4	21.4	16.5	11.5	6.5
Performance of BSF larvae						
f_p	%	0	32 ± 17	79 ± 17	80 ± 9	89 ± 6
t_p	day	–	23 ± 3	22 ± 1	19 ± 1	17 ± 1
Dry weight content	% WW	21 ± 1	23 ± 1	24 ± 1	27 ± 2	30 ± 2
Volatile solid	% DW	0.85	0.85	0.86	0.90	0.95
X_{max}	mg WW	34 ± 2	167 ± 11	185 ± 10	249.8 ± 19	308 ± 9
	mg DW	6.9 ± 0.4	38 ± 2	45 ± 2	67 ± 5	94 ± 3
μ_{max}	day ^{−1}	0.22 ± 0.01	0.54 ± 0.2	0.56 ± 0.01	0.58 ± 0.02	0.60 ± 0.01
Y	–	2.3 ± 0.4	0.76 ± 0.11	0.63 ± 0.06	0.59 ± 0.02	0.44 ± 0.04
M	kJ g ^{−1} DW	25.4	8.4	7.6	6.5	4.9
	day ^{−1}	0.36 ± 0.05	0.23 ± 0.01	0.15 ± 0.02	0.09 ± 0.01	0.08 ± 0.01
	kJ g ^{−1} DW day ^{−1}	4.0	2.6	1.6	1.1	0.9
	μW g ^{−1} WW	8.0	5.8	4.0	3.0	3.0
r^2	–	0.58	0.86	0.88	0.91	0.93
$q_{CO_2,max}$	day ^{−1}	0.86	0.64	0.53	0.44	0.35
	μW g ^{−1} WW	19.2	15.9	14.4	13.5	12.9
a_{max}	day ^{−1}	1.08	1.18	1.10	1.01	0.95
NGE_{max}	–	0.20	0.46	0.51	0.57	0.63
NGE_{avg}	–	0.09	0.22	0.31	0.39	0.48
$NGE_{avg} : NGE_{max}$	–	0.45	0.49	0.60	0.68	0.76

DW and WW represent dry weight and wet weight, respectively.

Nutritional analyses of substrates were performed by methods accredited for food analysis and the nutritional composition of degassed sludge may have deviated from measurements.

Error estimates indicate standard error (f_p and t_p) or standard error of mean (dry weight content, X_{max} , μ_{max} , Y , m) of 4 replicate BSF larval cultures.

X_{max} , μ_{max} , $q_{CO_2,max}$, a_{max} , Y , m , NGE_{max} and NGE_{avg} : NGE_{max} presented graphically in Supplementary information, Figs. S4–S6.

Day 8 and until prepupae formation was therefore characterised by declining r_X .

3.2. Cost of growth and maintenance costs

The rate of CO₂ production per larvae, Q_{CO_2} was maximal around at the same time as also r_X was maximal (Fig. 1, left panels). In Fig. 1 (central panels) specific CO₂ production rates, q_{CO_2} are compared to specific growth rates, μ estimated from Eq. (3). On all substrate mixtures did q_{CO_2} depend on μ , and the dependency was reasonably well described by a linear relationship as predicted by Eq. (5). The higher the CF content in mixed substrates, the higher was also the coefficient of determination, r^2 derived from linear regression analysis (Table 1), indicating that Eq. (5) provided the best description of the actual relationship between q_{CO_2} and μ when the CF content was high. The higher the content of DS in the substrate mixtures, the higher became also the cost of growth, Y and maintenance requirement, m ($p < 0.05$). Highest Y and m were seen in larvae grown on DS while lowest Y and m were observed in larvae grown on CF. Particularly m was affected by the proportion of DS in the substrate mixtures, while Y was remarkably high when DS was the sole substrate.

3.3. Net growth efficiency

Fig. 2 shows how NGE depended on the specific growth rate, μ on the different substrate mixtures. NGE depended strongly on μ , particularly when μ was lower than roughly 0.2 day^{−1}. NGE also depended on the ratio between DS and CF irrespectively of how

rapidly the BSF larvae were growing. Fig. 1, right panels show μ (Eq. (3)), specific CO₂ production rate, q_{CO_2} (Eq. (5)), specific assimilation rate, a (Eq. (4)), and NGE (Eq. (7)) against time of BSF larvae grown on the different substrate mixtures. All values were highest in small larvae. After Day 5, μ , q_{CO_2} and a began to decline. NGE remained close to its maximal value for another 3–5 days. The specific assimilation rate, a was almost the same across substrate mixtures the first days of the cultivations while NGE was considerably lower in larvae grown on DS compared to larvae fed also with CF. Maximal specific assimilation rate, a_{max} , maximal specific CO₂ production rate, $q_{CO_2,max}$, and maximal net growth efficiency, NGE_{max} were estimated from Eqs. (4), (5) and (8), respectively, using the maximal specific growth rates estimated for each substrate mixture (Table 1). Whereas a_{max} was almost independent of substrate composition, $q_{CO_2,max}$ was highest in larvae grown on 100% DS and lowest in larvae grown on 100% CF. Consequently, NGE_{max} was highest in larvae fed only with CF. Average net growth efficiencies, NGE_{avg} evaluated from Day 0 until the first prepupae emerged showed that 48% of the assimilated feed was converted into biomass in the BSF larvae fed only with CF while just 9% of the assimilated feed ended up as biomass in larvae fed only with DS.

4. Discussion

DS was a poor substrate for BSF larvae when supplied as the sole feed. Growth was slow, the larvae remained small, and no prepupae emerged. DS had also negative effects on larval size and prepupae formation when mixed with CF. Still, the ratio between DS

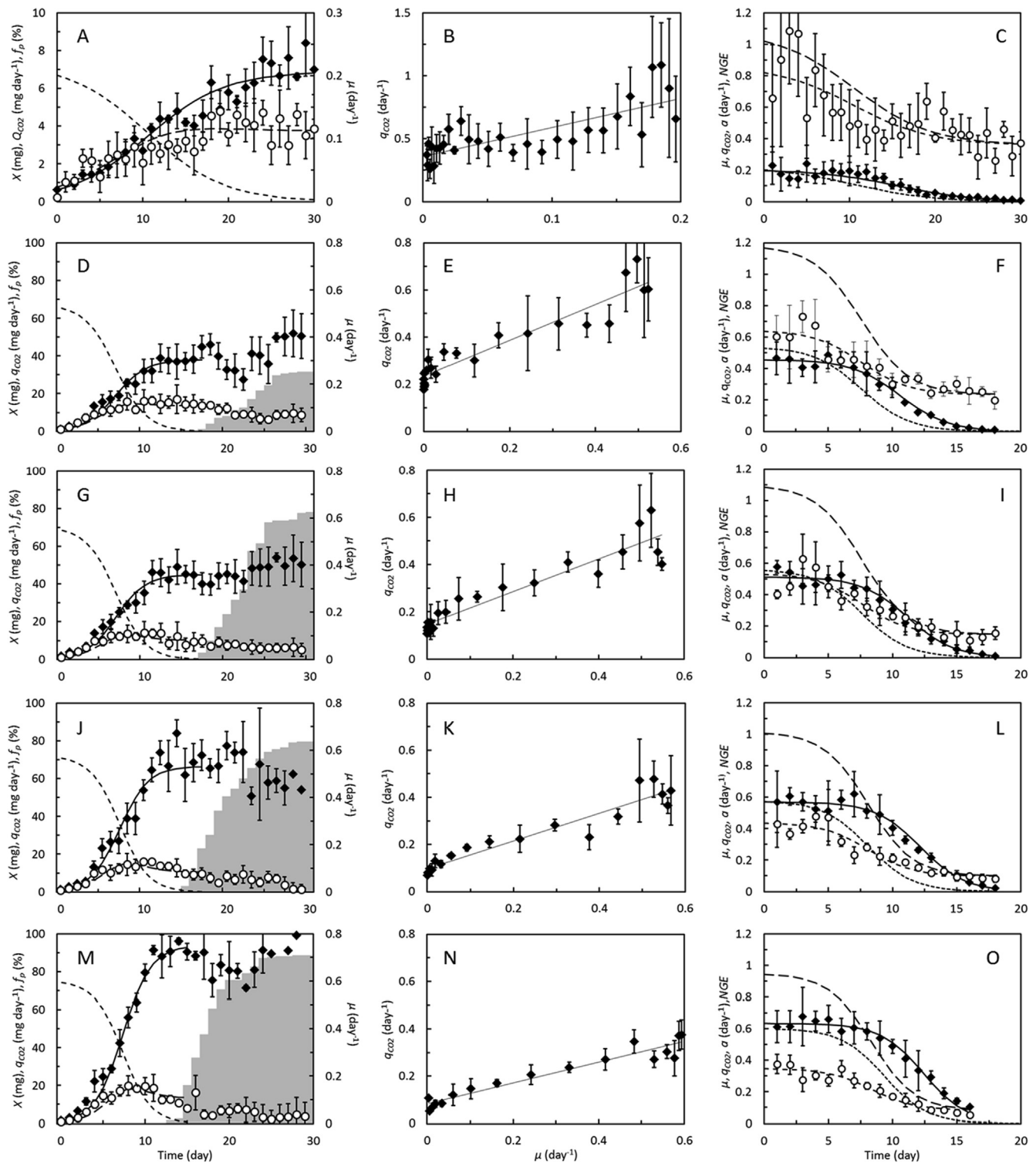


Fig. 1. BSF larvae grown on mixtures of degassed sludge (DS) and chicken feed (CF). A–C, 100% DS. D–F, 75% DS and 25% CF. G–I, 50% DS and 50% CF. J–L, 25% DS and 75% CF. M–O: 100% CF. Left panels show larval dry weight, X (●, solid curve, Eq. (1)), rate of biomass production, r_X (dotted curve, Eq. (2)), CO_2 production rate, Q_{CO_2} (○, dashed curve), and fraction of individuals transformed into prepupae, f_p (shaded area) for a period of 30 days. Central panels show specific CO_2 production rate, q_{CO_2} vs. specific growth rate, μ during growth phases. Regression lines represent Eq. (5). Right panels show specific growth rate (dotted curve, Eq. (3)), specific CO_2 production rate (○, dashed curve, Eqs. (3) and (5)), specific assimilation rate, a (dashed curve, Eq. (4)) and net growth efficiency, NGE (●, solid curve, Eq. (7)) during growth phases.

and CF had almost no effect on maximal specific assimilation rates, a_{max} and only minor effect on the maximal specific growth rate, μ_{max} , as long as CF was included in the substrate mixture. High specific rates of assimilation, a and growth, μ were, however, sustained for shorter periods of time when the sludge content was

high. Even though high contents of DS prolonged the time until prepupae were formed, t_p , the larvae grown on these substrate mixtures remained comparatively small. The cost of growth, Y and particularly the maintenance requirements, m increased when the ratio between DS and CF was high. Larger proportions of the

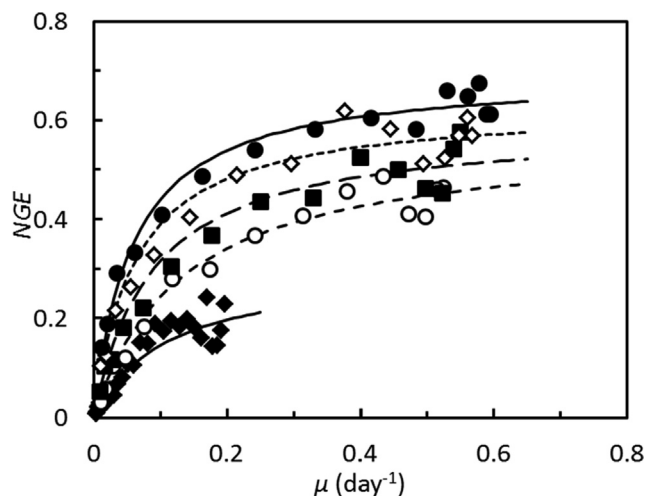


Fig. 2. Net growth efficiency, NGE of BSF larvae vs. specific growth rate, μ . Larvae grown on 100% degassed sludge (DS) ($\blacklozenge$, solid line), 75% DS and 25% chicken feed (CF) ($\circ$, dashed line), 50% DS and 50% CF ($\blacksquare$, dashed line), 25% DS and 75% CF ($\diamond$, dotted line), and 100% CF ($\bullet$, solid line). Curves determined by Eq. (8), costs of growth, Y and maintenance, m as shown in Table 1.

absorbed feed were therefore lost via respiration and, as a result, the net growth efficiency, NGE became low on substrate mixtures rich in DS.

The BSF cultures were in total supplied with 1.3 g (cultures grown solely on DS) or 2 g (remaining cultures) of dry substrate per larva, corresponding in amounts to 21–188 times the final dry weight of the larvae. Feed was therefore supplied in surplus and larval performances on the different mixtures of substrates can be compared based on the model outputs presented in Table 1. The reliability of these data depends on the ability of Eqs. (1) and (5) to actually describe growth and respiration. The growth experiments in Fig. 1 were started with 10 days old larvae and maximal sizes were obtained 12–15 days later. At this time, the mass of the larvae was 167–308 mg wet weight or 38–94 mg dry weight, except for larvae grown solely on DS. These results are in line with earlier studies, in which it has taken 15–40 days for BSF larvae to reach their maximal masses of 150–320 mg wet weight (Liu et al., 2017a; Barragán-Fonseca et al., 2018; Lalander et al., 2019). When the larvae formed prepupae they may have emptied their gut and started to rely solely on endogen metabolism, which is in line with the modest decreases in weight that occurred (Fig. 1). Until then growth curves were accurately described by the logistic function (Eq. (1)), although it describes growth as a continuous process not taking e.g. the transition between instars into consideration. Feed uptake in BSF larvae is restricted to instar 1–6 during the sigmoidal growth phase (Glăgorescu et al., 2019) and respiration rates were also seen to decrease as the larvae approached their maximal sizes (Fig. 1, left panels). During the growth phase, the specific CO_2 production rate, q_{CO_2} was close to linearly correlated to the specific growth rate, μ as predicted by Eq. (5), in particular when the CF content was high. Therefore, it seems that balanced growth is most closely accommodated when BSF larvae are reared on a high-quality substrate. The increasing DW contents, predominantly after Day 9 (Supplementary information, Fig. S1), probably reflect that the lipid content increased when the larvae approached the prepupal stage. Most of the biomass was, however produced before this point (Fig. 1, left panels).

The specific feed assimilation rate, a was almost the same on all substrate mixtures (Fig. 1, right panels and Table 1). Small BSF larvae assimilated a daily amount of feed corresponding roughly to their body size when measured in terms of carbon equivalents.

Unidentified compounds or conditions in the degassed sludge seemed to result in increased maintenance metabolism in the BSF larvae. The maintenance coefficient, m was 4 times larger in larvae grown only on DS compared to larvae grown only on CF (Table 1). The higher the content of DS in the substrate mixtures, the larger became the proportion of assimilated feed utilized for maintenance purposes and therefore unavailable for new biomass production. Since also the cost of growth, Y was higher on DS than on CF, NGE of BSF larvae grown on substrate mixtures rich in DS became low. Although the larval growth phase was extended by 6 days when the DS content was increased from 0% to 75%, this was insufficient to compensate for the decrease in growth rates. Prepupae produced on 75% DS ended up being only half the size of the prepupae produced on CF. The cost of growth in terms of energy ranging from $4.9 \text{ kJ g}^{-1} \text{ DW}$ on CF to $25.4 \text{ kJ g}^{-1} \text{ DW}$ on DS (Table 1) falls below and above, respectively, the cost of growth observed in pigs ($11.4\text{--}14.4 \text{ kJ g}^{-1} \text{ DW}$, van Milgen et al., 2001) and broiler chickens ($10.4\text{--}11.9 \text{ kJ g}^{-1} \text{ DW}$, Liu et al., 2017b). Specific maintenance requirements are expectedly related to the size of the animals (Riisgård, 1998). BSF larvae are comparatively small in size and their maintenance requirements of $0.9 \text{ kJ g}^{-1} \text{ DW day}^{-1}$ when grown on CF (Table 1) was also slightly higher than the $0.7\text{--}0.8 \text{ kJ g}^{-1} \text{ DW day}^{-1}$ that e.g. can be expected for a 0.1 kg broiler chicken (calculated from Liu et al., 2017b). The maintenance requirement of BSF larvae grown on DS ($4.0 \text{ kJ g}^{-1} \text{ DW day}^{-1}$) was much higher than what is seen in chickens.

It has been questioned if respiration rates measured in insects taken out of their normal living site and put into respiration chambers reflect the respiration rates of the undisturbed animals. Wightman (1981) found that so-called ‘flask effects’ resulted in respiration measurements below what should be expected from energy balances taking into account the ingested feed, the excreted faecal components, and growth. However, inconsistent energy balances may also be due e.g. to respiratory processes in some types of feed (McEvoy, 1985), and less than 10% discrepancy between predicted and measured respiration rates have been observed in insects reared in the laboratory. In this study, the larvae were gently taken out of their substrate immediately before respiration measurements, to avoid interference from microbial respiration, and transferred directly to the respiration chamber, in which air temperature was adjusted to 28°C and at atmospheric CO_2 partial pressure. The larvae spend only 5 min. in the respiration chamber to avoid starvation. CO_2 concentrations increased steadily during the measurement periods and the measured respiration rates were similar between equal-sized larvae from replicate cultures. This indicates that sufficient precautions were taken in order to avoid the larvae being severely disturbed during measurements. The metabolic rates (Table 1) can also be compared to earlier measurements in BSF larvae. The maximal specific CO_2 production rate $q_{CO_2, \max}$ for larvae grown on 100% CF corresponded to a specific power generation of $12.9 \mu\text{W mg}^{-1} \text{ WW}$ while the maintenance coefficient, m of these larvae corresponded to a specific rate of heat production of $3 \mu\text{W mg}^{-1} \text{ WW}$. These values are in almost perfect agreement with previous measurements of metabolic activity in BSF larvae, quantified by calorimetry (Glăgorescu et al., 2019). These authors found that the heat production decreased from $12 \mu\text{W mg}^{-1}$ of WW in rapidly growing BSF larvae below 10 mg WW in size to $2 \mu\text{W mg}^{-1}$ of wet weight in BSF larvae of 90 mg WW in stadium 6 when growth ceased. Such highly similar results suggest that the metabolic characteristics listed in Table 1 do represent general performances of BSF larvae reared on substrates of variable quality.

The low NGE on DS was expected. In an earlier study did BSF larvae digest 13% of the DS used as substrate while only 0.2% was converted into new biomass (Lalander et al., 2019), corresponding to a NGE of 0.02 (based on dry weight). This value is even lower than

the average net growth efficiency, NGE_{avg} of 0.09 (based on carbon equivalents) found in this study (Table 1). The assimilated feed from the substrate mixtures containing CF was converted 2–5 times more efficiently into new biomass, with NGE_{avg} values of 0.22–0.48. These values are well in line with previous studies, where 3.8–38% of the digested feed have been incorporated into new biomass (Diener et al., 2009; Abduh et al., 2018; Liu et al., 2018; Meneguz et al., 2018; Palma et al., 2019; Gold et al., 2020). The highest NGE value was found in small BSF larvae grown solely on CF. In these larvae, the maximal net growth efficiency, NGE_{max} attained a value of 0.63, which is comparable to what has been observed also in other insects (e.g. 0.44–0.89 in milkweed bugs, Chaplin and Chaplin, 1981) and invertebrates (e.g. 0.6 and 0.7 in the marine polychaetes *Nereis diversicolor* and *N. virens*, respectively, Nielsen et al., 1995). NGE_{max} marks the upper limit for feed conversion efficiency. Since NGE_{max} was lowered by DS (Table 1), also the potential for feed conversion into new biomass was reduced when DS was included in the substrate mixtures. None of the BSF cultures fulfilled their potential for feed conversion. The BSF larvae maintained high NGE for 8–10 days (Fig. 1, right panels) until they had gained 40–60% of their final weight. The remaining biomass was then produced at a NGE well below NGE_{max} and NGE_{avg} was therefore lower than NGE_{max} on all substrate mixtures (Table 1). High specific growth rates were maintained for relatively short periods of time on the substrate mixtures rich in DS, and the difference between NGE_{avg} and NGE_{max} was therefore highest when also the DS content was highest. BSF larvae grown solely on CF were reared with a NGE_{avg} corresponding to 76% of NGE_{max} . BSF larvae grown solely on DS were reared with a NGE_{avg} corresponding to only 45% of NGE_{max} . This means that there are two reasons for DS being a lower quality substrate than CF, both related to the metabolic performance of the BSF larvae: the potential for feed conversion into new biomass is substrate-specific and lowest on DS, and the BSF larvae meet their potential feed conversion to a lesser degree on DS.

5. Conclusion

The two substrates, degassed sludge (DS) and chicken feed (CF) strongly affected how assimilated feed was utilised by BSF larvae, causing large differences in feed conversion efficiency and growth. Although the larvae assimilated feed as rapidly from DS as from CF, they did not convert the feed from DS and CF into new biomass with the same efficiency. High rates of maintenance metabolism and increased cost of growth caused the larvae to utilise more feed for respiration purposes when DS was included in substrate mixtures, and less feed remained available for growth. DS also reduced the period of time the BSF larvae maintained growth at specific rates close to μ_{max} , and this increased the contribution of maintenance to the overall metabolism. The interplay between substrate quality and metabolic performance, here elucidated by a highly variable, substrate dependent NGE , may therefore have considerable consequences for the efficiency by which BSF larvae convert different substrates into new biomass.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2020.12.009>.

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